Santanu Mondal

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Summary

M.Tech graduate in **Network & Information Security** (Pondicherry University, 2026) with hands-on experience building blockchain applications, ML systems, and secure distributed software using **Python, JavaScript, Solidity, and Hyperledger Fabric**. Peer-reviewed publication and multiple end-to-end projects spanning smart contracts, zero-knowledge proofs, and NLP-based security tools. Seeking a **software engineering or security/blockchain engineering role**.

Skills

- **Programming:** Python, C, JavaScript, Java, Solidity
- **Blockchain & Web3:** Ethereum, Smart Contracts, Hyperledger Fabric, Hardhat, Ethers.js, Zero-Knowledge Proofs, Offline CBDC
- **Security & Compliance:** Cryptography, Security Auditing, Risk Management, ISO/IEC 27001, NIST, ISMS
- **Networking & Distributed Systems:** Socket Programming, TCP/UDP, Client–Server Architecture, Java RMI, Cisco Packet Tracer
- **AI & Data:** Machine Learning, NLP, scikit-learn, LightGBM, FastAPI, OpenCV, Pandas, NumPy
- **Web, Cloud & Tools:** HTML, CSS, AWS EC2, Git, GitHub, Linux, VS Code

Projects

- **MalURLX — NLP-Based Malicious URL Detection** [GitHub]
  Engineered an end-to-end ML system to detect malicious URLs from lexical features and character-level TF-IDF with a LightGBM classifier; served predictions through a FastAPI/Uvicorn REST API and shipped a companion Chrome extension for real-time browser protection.
- **Multiple Identity Management of IoT Devices — Blockchain** [GitHub]
  Designed a Hyperledger Fabric permissioned-blockchain system for managing multiple identities of IoT devices; implemented JavaScript chaincode (smart contracts) for device identity registration and asset transfer, built a client application on the Fabric Gateway SDK, and monitored the network with Hyperledger Explorer.
- **Decentralized Voting DApp** [GitHub]
  Developed an Ethereum voting smart contract (Solidity, Hardhat, Ethers.js) supporting multiple time-bound elections, one-vote-per-address enforcement, and real-time tallying; deployed and verified it on the Sepolia testnet via Alchemy RPC with event logging and admin access controls.
- **Third Eye — Edge AI Assistive Navigation System** [GitHub]
  Prototyped a fully on-device wearable navigation aid for visually impaired users on Arduino Nicla Vision at the ACM Winter School (IISc) Edge AI Hackathon; fused ToF distance sensing and IMU-based fall detection with an INT8-quantized vision model (10× smaller, 0.80 accuracy), streaming real-time alerts over UDP to a Flutter/Streamlit UI.
- **Remote Desktop — Real-Time Screen Sharing & Control** [GitHub]
  Built a Python remote-desktop tool that streams a host's screen and relays remote mouse/keyboard control; captured and JPEG-encoded frames with OpenCV, evolved the transport from TCP to TLS-secured and chunked-UDP streaming (~20 FPS), and replayed input events with pynput across multi-threaded clients.

Education

Pondicherry University — *M.Tech in Network & Information Security*, CGPA 9.13/10 2024–2026
Thesis: ZKP and PUF-Authenticated Dual-Offline CBDC Protocol for Resource-Constrained IoT Devices
South Asian University — *M.Sc in Computer Science* 2021–2023
Thesis: System for Managing Multiple Identities of IoT Devices using Blockchain
Asutosh College (University of Calcutta) — *B.Sc (Hons.) in Computer Science* 2018–2021

Selected Publications

1. S. Sharma, **S. Mondal**, S. Ishrat, "Managing Multiple Identities of IoT Devices Using Blockchain," *Soft Computing and Its Engineering Applications (icSoftComp 2023)*, CCIS 2031, Springer, 2024. DOI: 10.1007/978-3-031-53728-8_11
2. **S. Mondal** and T. Chithralekha, "Zero-Knowledge Proof (ZKP) Authentication for Offline CBDC Payment System Using IoT Devices," *Preprint*, arXiv:2603.03804.